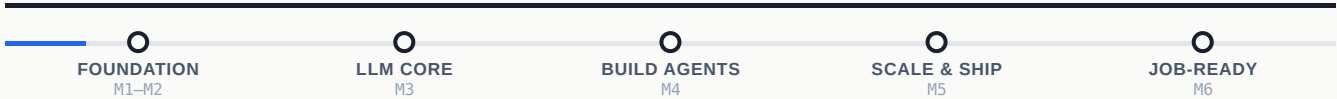


AI Engineer Career Roadmap



📊 CURRENT POSITION & STRENGTHS

Student with a genuinely strong ML base — most candidates targeting AI Engineer roles start *without* this. You already cover the classic ML lifecycle (prep → train → evaluate → deploy) plus frontend, which is rare and valuable for building full-stack AI products.

- ✓ Python & ML Libs
- ✓ Feature Engineering
- ✓ Model Evaluation
- ✓ Preprocessing
- ✓ Deployment
- ✓ Frontend Dev

⚡ SKILL GAP ANALYSIS

GAP AREA	WHY IT MATTERS	PRIORITY
LLM APIs & Prompting	Core daily tool of an AI Engineer	HIGH
RAG Pipelines	Most hired-for AI Engineer task in 2026	HIGH
Agentic Frameworks	LangGraph/CrewAI now standard in JDs	HIGH
Vector DBs	Required for retrieval & memory systems	MED
LLMOps & Eval	Differentiates hire-ready from hobbyist	MED

🌟 RECOMMENDED LEARNING PLAN

- 01 · LLM Foundations** MONTH 1-2
OpenAI/Anthropic APIs, prompt engineering, tokenization, embeddings, function calling, structured outputs.
- 02 · RAG & Retrieval** MONTH 3
Vector DBs (Pinecone/Chroma), chunking strategies, hybrid search, reranking, evaluation (RAGAS).
- 03 · Agentic Systems** MONTH 4
LangGraph/CrewAI, tool calling, multi-agent orchestration, memory, guardrails.
- 04 · Production & LLMOps** MONTH 5-6
FastAPI serving, Docker, observability (LangSmith), cost/latency optimization, fine-tuning basics (LoRA).

📁 SUGGESTED PROJECTS

- ① RAG-powered docs assistant**
Full-stack app: your frontend skills + a retrieval backend over real documents. Deploy live.
- ② Autonomous multi-agent workflow**
Agents that plan, call tools, and self-correct — e.g. research-and-report agent crew.
- ③ End-to-end ML+LLM pipeline**
Combine prior ML skills (preprocessing/eval) with an LLM layer; deploy with monitoring.
- ④ Open-source contribution**
One merged PR to a known AI tooling repo (LangChain, LlamaIndex) — strong signal to recruiters.

🌐 NETWORKING STRATEGY

Build in Public Post weekly progress + project breakdowns on LinkedIn/X with #buildinpublic.	Targeted Outreach DM 5 AI engineers/week at target companies; ask 1 specific question, not "any advice?"	Communities Active in 2-3 Discords/Slacks (LangChain, local AI meetups, Kaggle).	Visible Proof Pin GitHub + live demos to LinkedIn featured section; write 1 technical post/month.
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📅 MONTHLY MILESTONES

MONTH	FOCUS	DELIVERABLE	PROOF POINT
M1	LLM API fundamentals + prompting	5 small prompt-engineering scripts/notebooks	GitHub repo live
M2	Embeddings & vector search	Mini semantic search tool	Demo on LinkedIn
M3	RAG pipeline mastery	Project ① — RAG docs assistant	Deployed + shared publicly
M4	Agentic frameworks	Project ② — multi-agent workflow	Demo video + writeup
M5	Production hardening (Docker, monitoring)	Project ③ — full ML+LLM pipeline	Live URL + metrics dashboard
M6	Job-readiness sprint	Portfolio site + resume + 1 OSS PR	Applying to AI Engineer roles

⚡ IMMEDIATE NEXT ACTIONS (THIS WEEK)

- 1
Get OpenAI + Anthropic API keys; run your first 3 prompt-engineering scripts.
- 2
Create a public GitHub repo named "ai-engineer-roadmap" and commit daily.
- 3
Post your starting point + 6-month plan on LinkedIn to create accountability.
- 4
Pick Project ① and scope its v1 feature list before Month 2 begins.